Felipe Pérez Silva

Ph.D. Student in Mathematics, George Mason University

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Research Interests

Nonlinear analysis of partial differential equations; numerical analysis and optimal control of PDEs; convex duality methods; finite element methods. Specific interests include existence and multiplicity of solutions via variational techniques (Mountain Pass and linking theorems, fixed-point methods, sub- and super-solutions), the p-Laplace operator under homogeneous and nonhomogeneous Dirichlet boundary conditions, obstacle problems, and a posteriori error analysis.

Education

2024 – present **Ph.D. in Mathematics**
George Mason University, Fairfax, VA, USA
Advisor: Dr. Harbir Antil

2017 – 2023 **Professional Degree in Mathematical Engineering** (*Ingeniería Civil Matemática*)
Universidad Técnica Federico Santa María, Valparaíso, Chile
Six-year integrated program. B.Sc. in Mathematical Engineering awarded in 2020 (years 1–4). The M.Sc. below was pursued concurrently during years 5–6, with a single joint thesis defense.

2021 – 2023 **M.Sc. in Mathematics**
Universidad Técnica Federico Santa María, Valparaíso, Chile
Advisor: Dr. Leonelo Iturriaga
Thesis (joint defense with Mathematical Engineering): "Existence and multiplicity of positive solutions for a quasilinear problem with nonhomogeneous Dirichlet condition."

Publications

Submitted / Under Review

1. H. Antil, R. Löhner, and **F. Pérez**, "Optimal boundary control of diffusion on graphs via linear programming," *Submitted, 2025*. arXiv:2511.03129.
2. L. Iturriaga and **F. Pérez**, "Existence and multiplicity of positive solutions for a quasilinear problem with nonhomogeneous Dirichlet condition," *Submitted, 2025*.

In Preparation

3. H. Antil, A. Kaltenbach, **F. Pérez**, T. Prieto, and V. Lagos, "Convex duality and guaranteed a posteriori error control for curl–curl Maxwell problems with pointwise field bounds," *In preparation, 2026*.
4. H. Antil, K. L. A. Kirk, and **F. Pérez**, "Duality framework for flux constrained flow: analysis and numerics," *In preparation, 2026*.
5. H. Antil and **F. Pérez**, "Duality framework for the double obstacle problem: analysis and numerics," *In preparation, 2026*.

Research Experience

2024 – present **Graduate Research Assistant**
George Mason University · PI: Dr. Harbir Antil
Research on convex-duality methods and a posteriori error control for constrained PDE problems, including obstacle problems, flux-constrained flows, and curl–curl Maxwell systems with pointwise field bounds. Optimal boundary control of diffusion on graphs via linear programming.

2021 – 2023 **Research Assistant**
Universidad Técnica Federico Santa María · PI: Dr. Leonelo Iturriaga
Research on existence and multiplicity of positive solutions for quasilinear elliptic problems involving the p-Laplacian under nonhomogeneous Dirichlet conditions. Variational and sub/super-solution methods.

Summer 2020 **Software Engineering Intern**
Inria Chile, Santiago, Chile
Optimized Django REST Framework backend queries to reduce database load. Implemented React and RxJS frontend improvements for application performance.

Teaching Experience

Instructor of Record

Spring 2024 *Universidad Técnica Federico Santa María*
Term Professor (sole instructor): Analytic Geometry; Calculus I; Calculus II.

Graduate Teaching Assistant

2024 – present *George Mason University*
Calculus I and Calculus II: led recitation sections, held office hours, graded exams and assignments.

Laboratory & Teaching Assistant

2023 *Universidad Técnica Federico Santa María*
Laboratory Assistant: Differential Calculus; Algebra & Geometry.

2018 – 2022 *Universidad Técnica Federico Santa María*
Teaching Assistant for Differential Calculus, Integral Calculus, Algebra & Geometry, Linear Algebra, and Partial Differential Equations.

Talks & Presentations

2025 **Finite Element Circus**, George Mason University.
Duality Framework for Flux Constrained Flow: Analysis and Numerics.

2025 **Seminar on Calculus of Variations**, Georgetown University. Invited speaker.

2024 **Seminar on Control of Partial Differential Equations**, George Mason University. Presenter and attendee; talks on the Fenchel conjugate and primal–dual formulations for PDE-constrained problems.

2021	**Seminar on Partial Differential Equations** (Doctoral Program), Universidad Técnica Federico Santa María. Presenter on minimax methods (Mountain Pass and linking theorems), the Nemytskii operator, and the deformation lemma.

Workshops & Posters

2026	**ICERM**, Brown University. *Simulation-Based Optimization with Applications*. Poster presentation.
2025	**IMSI**, University of Chicago. *Optimal Control and Decision Making Under Uncertainty for Digital Twins*. Poster presentation.

Skills

Programming	Python (NumPy, SciPy), MATLAB, C++, LaTeX.
Numerical PDE	FEniCS, NGSolve.
Web (legacy)	Django REST Framework, React, RxJS (from Inria internship).
Languages	Spanish (native); English (professional working proficiency).